Some Details

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October 1, 2025

1 Normalization of Polarization Vectors

First, we give the definition of polarization vectors which is written in terms of adjoint representation

$$\varepsilon_a^\mu = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p^J\rangle(\sigma_a)^I{}_J}{4(p\cdot q)} \tag{1}$$

here a runs from 1 to 3. We can use antisymmetric tensor to rewrite this equation,like

$$\varepsilon_a^\mu = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p_K\rangle\epsilon^{JK}(\sigma_a)^I{}_J}{4(p\cdot q)} \tag{2}$$

here we choose the left-multiplication convention.

Then, a 2×2 symmetric matrix can be defined like

$$(\sigma_a)^{IJ} = \epsilon^{JK}(\sigma_a)^I{}_K. \tag{3}$$

This means that the original polarization vector can be understood as

$$\begin{aligned}\varepsilon_a^\mu &= -\frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p_J\rangle}{4(p\cdot q)}(\sigma_a)^{IJ} \\ &= -\varepsilon_{IJ}^\mu(\sigma_a)^{IJ}\end{aligned} \tag{4}$$

here I, J are fundamental little group indices, runs from 1 to 2.

And we know that the reasonable polarization vector for physical state should be symmetric with little group indices, it means that we need to symmetrize the indices I and J as $(IJ) = \frac{1}{2}(IJ + JI)$

Now, we want to compute the normalization for ε_{IJ}^μ, and after multiplying the symmetric matrix, it should go back to the normalization

$$\varepsilon_a^\mu\varepsilon_{b,\mu} = -\delta_{ab} \tag{5}$$

The method to compute the normalization is just multiplying two polarization vectors directly, like

$$\varepsilon_{(IJ)}^\mu\varepsilon_{(KL),\mu} \tag{6}$$